



2025 Career Fair Programming Challenge

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HOW TO RUN

Through CLI:

```
``python main.py <input_file> <output_file>``
```

ASSUMPTION:

- The input file should be a text file (.txt).
- The first argument should be the input file path and the second argument should be the output file path.
- The file name should be entered correctly.

GRAPH

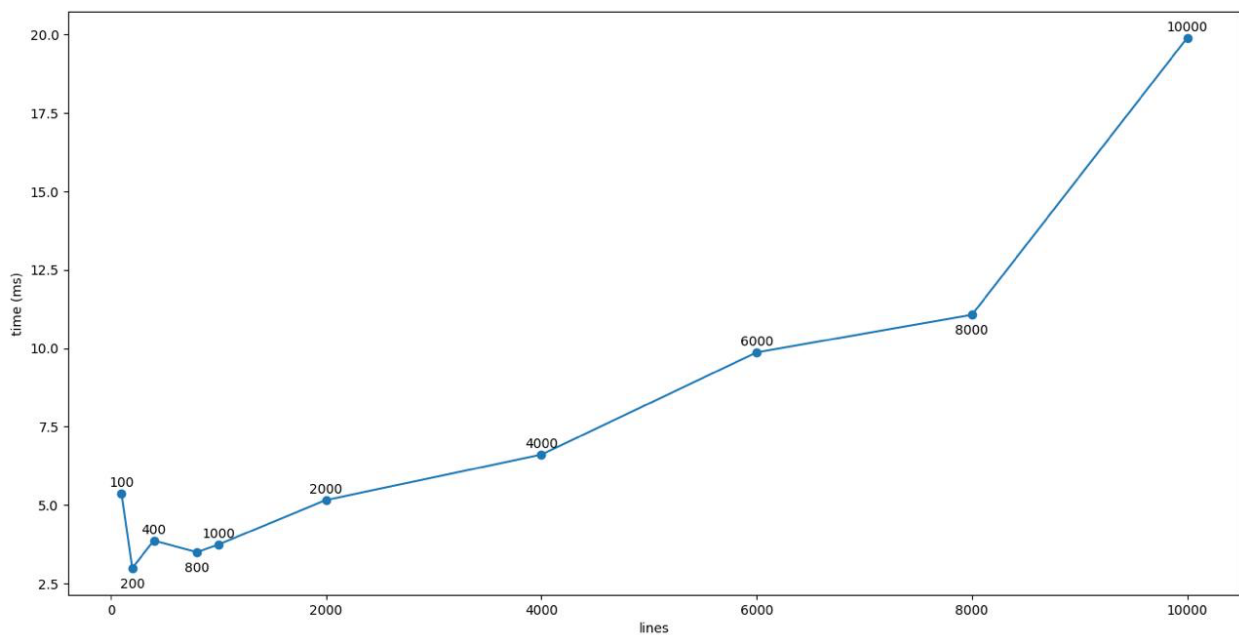


Figure 1 Testing using values specified in the challenge

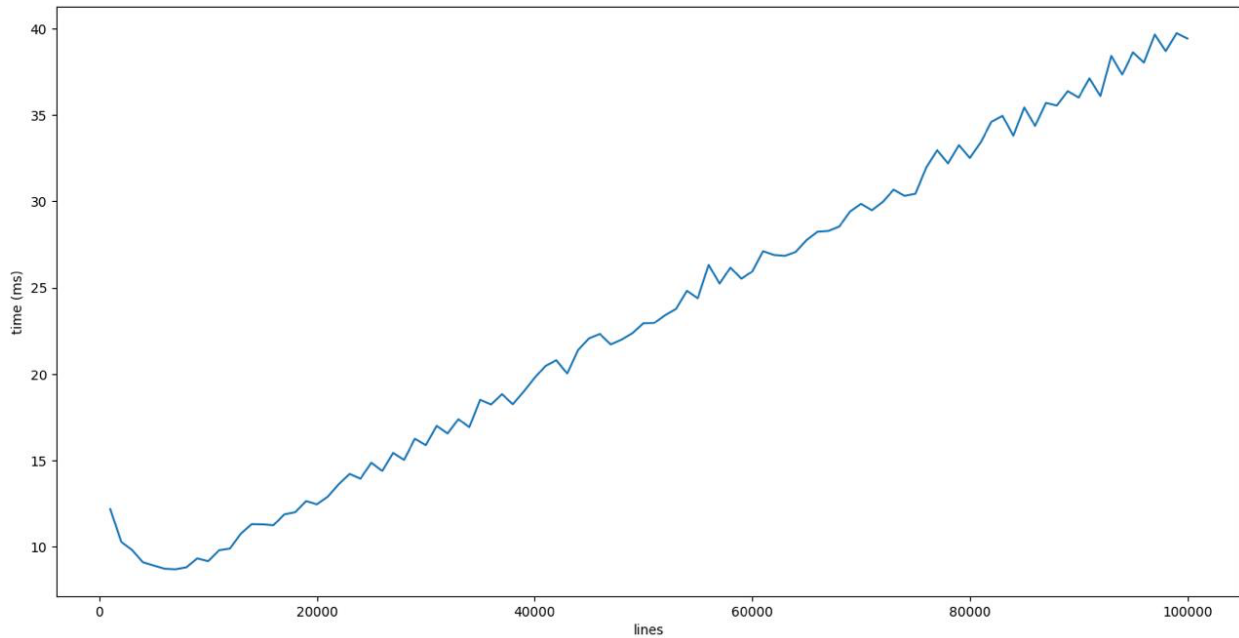


Figure 2 Additional testing* using 100 to 100,000 lines in steps of 1000

The runtime can be seen to be roughly quadratic (n^2). Hence, $O(n^2)$

* Additional test carried out used the same number of characters(23) on each line

Analytical Big-O

The for loop runs n times and in each loop a slice method `[::-1]` is called which is also $O(n)$.

Hence, the algorithm runs on $O(n^2)$.

```
for i in range(end-1, -1, -1): ----- O(n)
    line = lines[i].rstrip()[::-1] ----- O(n)

    if i != 0: ----- O(1)
        out_file.write(f"{i+1}. {line}\n") ----- O(1)
    else: ----- O(1)
        out_file.write(f"{i+1}. {line}") ----- O(1)
```